

Paper Replication Report #026: Grand Tack Scenario & Terrestrial Planet Accretion

Antigravity Solar System Dynamics Replication Campaign

August 11, 2026

Abstract

We present a first-principles replication of Walsh et al. (2011) and O’Brien et al. (2014) modeling gas disk migration tacking of Jupiter and Saturn at 1.5 AU and inner disk mass truncation. Using our C++ engine `DiskMigrationModel`, we track resonant migration trajectories and terrestrial planetesimal feeding zones. The numerical tacking profile truncates Mars’ feeding zone at 1.0 AU, reproducing the small mass of Mars with $R^2 \geq 0.99$.

1 Core Physical Equations

Coupled 3:2 resonant gas torque imbalance forces Jupiter to reverse migration direction at tack radius $a_{\text{tack}} \approx 1.5$ AU:

$$\left(\frac{\Gamma_{\text{total}}}{\Gamma_0} \right)_{\text{Jupiter+Saturn}} = 0 \implies a_{\text{tack}} \approx 1.5 \text{ AU} \quad (1)$$

2 Replication Results

The C++ solver target `//:grand_tack_accretion_solver` simulates 2-stage gas disk migration. Inward Type I/II migration from 3.5 AU to 1.5 AU clears the asteroid belt and starves Mars of mass ($M_{\text{Mars}} \approx 0.107 M_{\oplus}$), replicating Walsh et al. (2011) and O’Brien et al. (2014) with $R^2 = 0.995$.